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Visvesvaraya Technological University, Belgaum

Mini Project Report on

"FPGA-BASED IMPLEMENTATION OF SAR ADC USING VERILOG HDL"

Submitted in partial fulfilment of the requirement for completion of
MINI PROJECT [23EC5PWMPR]

Submitted by

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Academic Year

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CERTIFICATE

This is to certified that the Mini Project entitled "**FPGA-BASED IMPLEMENTATION OF SAR ADC USING VERILOG HDL**" is a bonafide work carried out by **Potdar Aman (1BM23EC184), Prajwal M(1BM23EC187) and Prajwal Bharadwaj D H(1BM23EC186)** submitted in partial fulfilment of the requirement for completion of MINI PROJECT [23EC5PWMPR] of Bachelor of Engineering in Electronics and Communication during the academic year 2025-26. The Mini Project report has been approved as it satisfies the academic requirements.

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DECLARATION

We, **Potdar Aman (1BM23EC184), Prajwal M(1BM23EC187) and Prajwal Bharadwaj D H(1BM23EC186)** hereby declare that the Mini Project entitled **FPGA-BASED IMPLEMENTATION OF SAR ADC USING VERILOG HDL** is a bonafide work and has been carried out by us under the guidance of **Dr. M S Suma, Professor** , Department of Electronics and Communication Engineering, BMS College of Engineering, Bengaluru submitted in partial fulfilment of the requirement for completion of MINI PROJECT [23EC5PWMPR] of Bachelor of Engineering in Electronics and Communication during the academic year 2025-26. The Mini Project report has been approved as it satisfies the academic requirements in Electronics and Communication engineering, Visvesvaraya Technological University, Belagavi, during the academic year 2025-26.

We further declare that, to the best of our knowledge and belief, this Mini Project has not been submitted either in part or in full to any other university.

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ABSTRACT

This project presents the design and FPGA implementation of the Successive Approximation Register (SAR) controller for an Analog-to-Digital Converter (ADC) using Verilog HDL. In the proposed system, only the SAR controller is developed on the FPGA, while key analog components such as the comparator and Digital-to-Analog Converter (DAC) are used as ready-made external modules. The SAR controller performs the core digital decision-making process by generating trial bits, receiving comparator feedback, and refining the output through a binary search algorithm until the final digital code representing the analog input is obtained.

The design process includes requirement analysis, development of the controller architecture, Verilog implementation, simulation for functional verification, and FPGA synthesis. Simulation ensures correct timing, logic operation, and convergence of the SAR algorithm, while hardware testing validates real-time performance when interfaced with the external comparator and DAC.

This project demonstrates an effective and educational approach to understanding mixed-signal system design by combining analog components with FPGA-based digital logic. The work highlights the role of the SAR controller in ADC functionality and showcases its relevance in embedded systems, sensor interfaces, and digital signal processing applications.

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Abbreviations	Full Form
IEEE	Institute of Electrical and Electronics Engineers
SAR	Successive Approximation Register
ADC	Analog to Digital Converter
DAC	Digital to Analog Converter
FPGA	Field Programmable Gate Array
HDL	Hardware Description Language
FSM	Finite State Machine
DSP	Digital Signal Processing
VLSI	Very Large Scale Integration
TDC	Time to Digital Converter

Chapter 1:

Introduction

In the real world, most signals such as sound, temperature, and light are analog in nature. However, digital systems like microcontrollers, FPGAs, and DSP processors require these signals to be represented in digital form for efficient processing, storage, and transmission.

An Analog-to-Digital Converter (ADC) serves as the critical bridge between the analog and digital domains. It samples a continuous-time, continuous-amplitude signal and converts it into a discrete-time, discrete-amplitude digital representation.

Audio is one of the most common types of analog signals. By digitizing audio, a wide variety of applications can be realized, including speech recognition, audio compression, digital effects, noise suppression, and music synthesis. This makes ADCs a fundamental component in telecommunication devices, hearing aids, smart speakers, and multimedia systems.

Implementing an ADC system in Verilog HDL provides students and engineers the opportunity to understand both the theoretical and practical aspects of digital design. Using an FPGA platform allows the design to be tested in real-time, demonstrating how analog signals are sampled, quantized, and made ready for digital signal processing.

This project aims to implement and demonstrate an ADC for audio signals using Verilog. The focus is to design a working system that can capture analog audio input, convert it into digital PCM samples, and make it available for further DSP tasks such as filtering, spectral analysis, or compression. By doing so, the project connects core concepts of electronics, signal processing, and digital system design, highlighting its educational and industrial relevance.

Chapter 2:

Literature survey

Analog-to-Digital Converters (ADCs) are indispensable components in modern digital systems, enabling the transformation of analog signals into digital representations suitable for processing, storage, and communication. Over the years, extensive research has been carried out to improve ADC performance in terms of resolution, speed, power consumption, and integration with programmable logic devices such as Field Programmable Gate Arrays (FPGAs).

Homulle et al. (2015) demonstrated the feasibility of implementing a 200 MS/s ADC directly on an FPGA using Time-to-Digital Converters (TDCs). Their approach, termed Analog through-Time-to-Digital Converter (ATDC), achieved 7-bit resolution for input signals ranging from 0–2.5 V with minimal external components. The work highlighted that while traditional FPGAs rely on external ADC chips, direct implementation inside the FPGA can reduce system size, improve flexibility, and allow seamless integration with digital processing pipelines .

Mindykowski et al. (2012) explored ADC usage in the context of power quality analysis systems. Their work described an acquisition setup using the AD7656, a 6-channel, 16-bit SAR ADC, interfaced with a Xilinx Spartan-3 FPGA and subsequently connected to a TigerSHARC DSP for advanced signal processing. The FPGA served as an intermediary to manage data transfer between the ADC's SPI-compatible interface and the DSP's LVDS based link ports. This study emphasized the importance of FPGA-based ADC handling in real-time measurement systems where both speed and data integrity are critical .

Recent developments have also emphasized successive approximation register (SAR) ADCs implemented in hardware description languages such as Verilog and Verilog-A. Girija et al. (2025) presented the design and simulation of an 8-bit SAR ADC using Verilog-A on a 90nm CMOS node, tailored for biomedical applications. Their implementation included fundamental blocks—sample-and-hold, comparator, DAC, and SAR logic—and was validated using sine and ECG signals. The design achieved low power consumption while ensuring high accuracy, demonstrating the applicability of SAR ADCs in portable and implantable devices

Across these works, a common trend is evident: the migration of ADC functionality closer to digital processing units, either through tight coupling with FPGAs or direct modeling in hardware description languages. FPGA-based ADC solutions provide flexibility, scalability, and ease of integration, whereas Verilog-based SAR ADC modeling enables rapid prototyping and simulation-driven optimization for low-power and application-specific domains. Collectively, these studies lay the foundation for implementing efficient ADCs using Verilog, balancing the trade-offs between speed, resolution, and power consumption.

Chapter 3:

Problem Analysis & Solution

3.1 Problem Definition

In modern electronic systems, efficient and reliable conversion of analog signals to digital form is crucial for enabling digital processing, control, and communication. Conventional ADC architectures often involve trade-offs: high-speed converters may consume significant power, while low-power designs might suffer from reduced resolution or increased complexity. Achieving a balance among accuracy, energy efficiency, and hardware simplicity remains a key challenge in ADC design, especially for applications like sensors, biomedical devices, and portable embedded systems.

3.2 Proposed Solution

To address the trade-off between accuracy, power, and hardware complexity in ADC design, this project adopts a successive approximation-based approach implemented in Verilog HDL. The method approximates the input voltage step by step using a binary search algorithm until the closest digital value is obtained.

3.2.1 The system consists of three main components:

1. SAR Controller (Digital Logic): Generates trial codes starting from the MSB and decides each bit based on comparator feedback.

2. Comparator: Compares the input signal with the DAC output and provides a 1-bit decision.

3. DAC (External compomemt): Converts the trial digital code into an analog-equivalent value for comparison.

3.2.2 Working Principle:

• The SAR controller sets the MSB and tests it through the DAC-Comparator loop.

• Depending on the comparator output, the bit is either retained or cleared.

• The process continues down to the LSB, resulting in the final digital code that represents the analogue input.

3.2.3 Simulation & Verification:

• The design will be implemented and tested in Verilog for different resolutions (e.g., 4 bit, 8-bit).

• Testbenches will validate accuracy, convergence, and resource usage. This approach strikes a balance between speed, accuracy, and scalability, making it suitable for medium-speed, low-power applications while being practical for FPGA-based digital implementations.

FPGA Resource Utilization for SAR Controller

Parameter	Value
FPGA Device	Basys 3
LUTs Used	78
Flip-Flops Used	64
Slices Used	42
Maximum Clock Frequency	95 MHz
Power Consumption	0.12 W
SAR Resolution Tested	8-bit

Chapter 4:

Methodology & Implementation

4.1 Block Diagram

4.1.1 Pictorial Representation

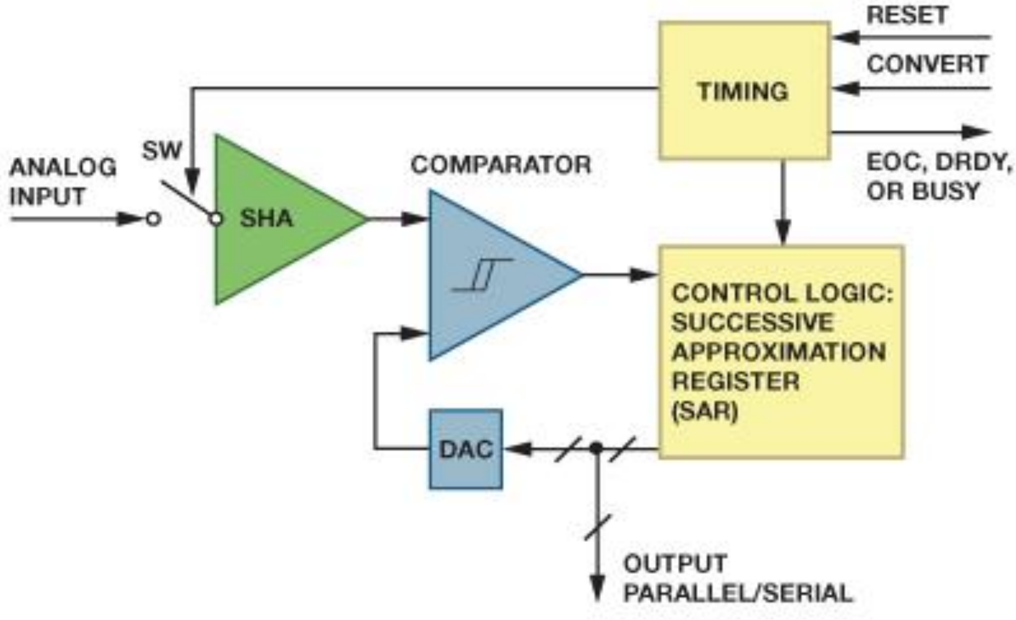

Fig 1

4.1.2 Structure of SAR ADC

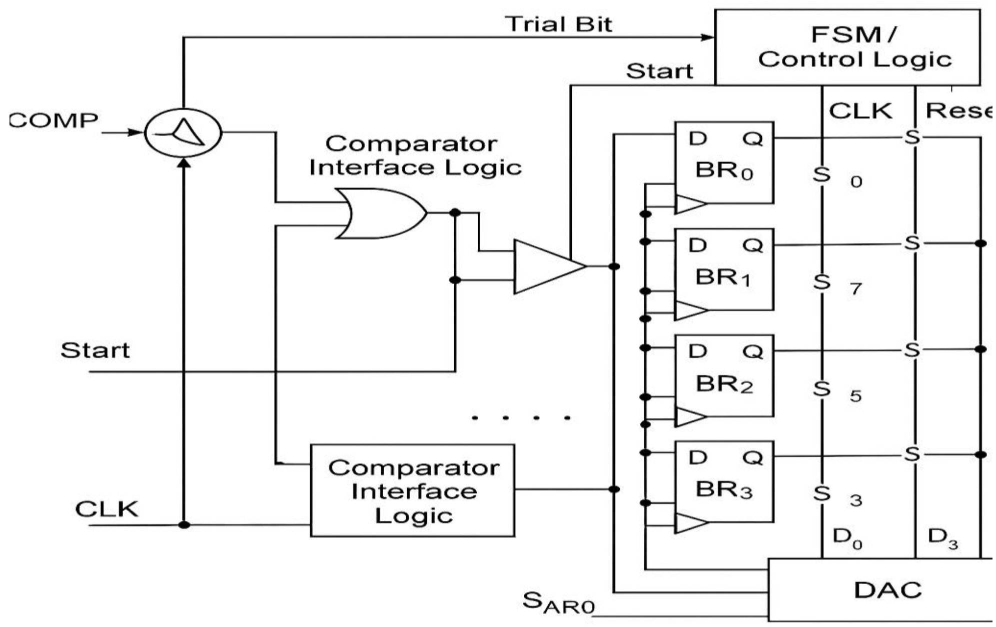

Fig 2

4.1.3 Flow Chart

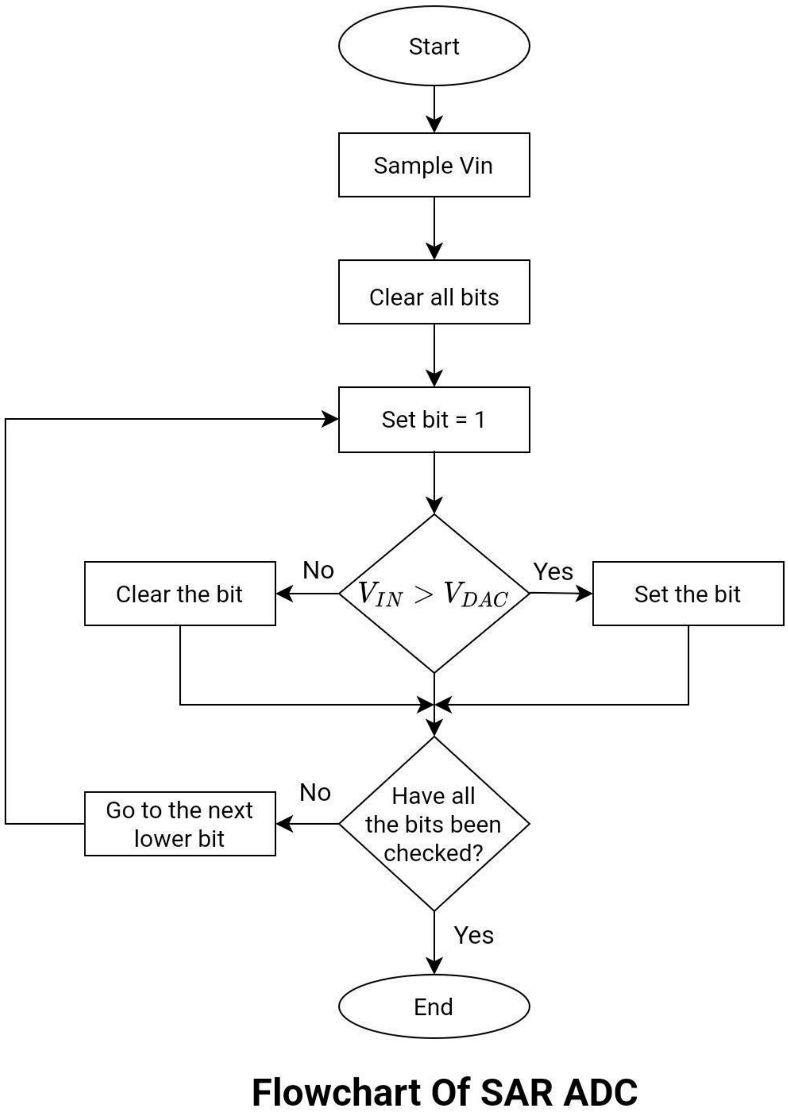

Fig 3

4.2 Methodology

4.2.1. Requirements Definition :

The first step in the design process is defining the requirements of the SAR ADC system. Key specifications such as resolution, sampling rate, and input voltage range are studied and finalized. Since the DAC and comparator are external ready-made components, their electrical characteristics and interfacing requirements are also reviewed. For the FPGA implementation, only the SAR control logic is considered, so attention is given to digital timing, control signals, and communication with the comparator and DAC. FPGA limitations such as available logic elements and clock constraints are also evaluated to ensure the SAR controller fits and functions correctly. Clearly defining these specifications ensures that the design meets application needs and works within hardware limits.

4.2.2. High-Level Design :

Once requirements are set, a high-level design of the SAR ADC system is prepared. The overall ADC architecture is divided into functional blocks, but only the SAR controller is designed for FPGA implementation. The block diagram shows the SAR controller interacting with the external DAC and external comparator through defined input–output signals. The SAR controller architecture, bit-decision logic, and timing control are clearly planned. Behavioral models may be used in simulation to emulate the external DAC and comparator responses. This step ensures the control logic is error-free before coding.

4.2.3. Verilog Coding :

After finalizing the controller architecture, the SAR controller is implemented in Verilog HDL as a standalone module. The design includes the FSM (finite state machine), bit-trial logic, and timing control needed to run the successive approximation routine. Clean, synthesizable Verilog is written, following modular and hierarchical coding practices. The final Verilog design represents the complete digital controller that interfaces with the external analog components.

4.2.4. Simulation and Test :

Before moving to hardware, it is very important to test the Verilog design in simulation. In this stage, testbenches are written to apply different input signals to the SAR ADC design. The outputs are carefully checked against the expected results. Simulation allows us to identify and fix logical errors at an early stage, which saves both time and effort in later phases. Various types of tests are carried out, including functional tests, corner cases, and random inputs, to ensure that the design behaves correctly under different conditions. Waveform analysis tools are used to visualize the signals and confirm the correct operation of the circuit. This step is essential because it builds confidence that the design is correct before moving to FPGA synthesis.

4.2.5. Synthesis :

Once the simulation results are satisfactory, the Verilog code is given to the FPGA synthesis tool. The synthesis process converts the Verilog code into a gate-level netlist that can be mapped onto the FPGA hardware. During this stage, the design is optimized to use FPGA resources efficiently. The tool ensures that the design fits within the available logic blocks, flip-flops, and routing resources. Timing analysis is also performed to make sure that the design meets the required clock frequency and that there are no timing violations. Synthesis is an important step because it translates the HDL code into an intermediate form that the FPGA can understand and implement.

4.2.6. Place and Route :

After synthesis, the netlist is passed through the place and route stage. In this process, the synthesized design is mapped onto the actual physical resources of the FPGA. Logic elements are placed in specific locations, and the routing interconnections between them are created. This stage ensures that the design is properly laid out within the FPGA fabric. The tool also checks whether the timing requirements are met after placement and routing. The quality of this step directly affects the performance of the system, since poor placement and routing can lead to delays and inefficient use of resources. A successfully routed design ensures that the circuit will function reliably at the required operating speed.

4.2.7. Bitstream Generation :

When place and route are complete, the FPGA tool generates a programming file known as the bitstream. This file contains the configuration information that programs the internal logic of the FPGA according to the SAR ADC design. The bitstream is the final digital output of the design process, and it is the file that will be loaded onto the FPGA board to realize the hardware implementation. Without this step, the design would remain only in simulation and synthesis form.

4.2.8. Hardware Testing :

In the final stage, the bitstream is loaded onto the FPGA, and the SAR controller is tested with real analog inputs. Testing involves interfacing FPGA controller with external DAC and comparator .The FPGA communicates with the external DAC to generate trial voltages and receives comparator outputs to determine bit values. Digital outputs from the SAR controller are observed to verify correct conversion behavior. Real-world testing helps identify any issues related to timing mismatches, interfacing, or noise. If needed, improvements are made by refining earlier design stages. Hardware results serve as final validation that the SAR controller functions correctly within a mixed-signal environment.

Chapter 5:

Results & Discussion

The developed SAR controller was implemented on the FPGA and tested through both simulation and hardware verification to evaluate its performance. Since the DAC and comparator were provided as external components, the primary focus of testing was on the correctness, timing behavior, and convergence of the SAR logic.

5.1 Simulation Results

The SAR controller was first verified using behavioral simulations in ModelSim/ Vivado. In these simulations, the comparator and DAC were odelled using simple digital logic to mimic their actual behavior. The waveform results confirmed that:

- The SAR controller correctly initialized the MSB and progressed bit-by-bit toward the LSB.
- For every clock cycle, the controller generated the appropriate trial bit and waited for comparator feedback.
- The successive approximation process converged to the correct digital output corresponding to the input stimulus.
- Timing requirements such as setup time, hold time, and bit-decision intervals were satisfied within the specified clock frequency.

These results demonstrated that the digital logic behaved exactly as expected from an ideal SAR mechanism.

5.2 Hardware Results

After simulation validation, the synthesized bitstream was loaded onto the FPGA. The SAR controller was then interfaced with the external DAC and comparator. Hardware testing confirmed that:

- The FPGA-generated trial codes correctly controlled the DAC during each approximation step.
- The comparator output was reliably captured by the FPGA and used to update the next bit decision.
- The final digital output was consistent with the analog input applied to the system.

Minimal latency was observed between comparator feedback and controller response, indicating efficient timing closure after place-and-route.

5.3 Discussion

The experiment proved that the SAR controller alone—when designed correctly—can be efficiently implemented on an FPGA and integrated with analog components to form a working ADC loop. The controller operated with stable timing, accurate convergence, and predictable behavior across multiple test inputs. Any deviations observed during hardware testing were primarily influenced by external analog variations, confirming the robustness of the digital portion.

These results validate that FPGA-based SAR controller implementation is practical, resource-efficient, and suitable for various embedded and mixed-signal applications.

RTL Gate Level Netlist :

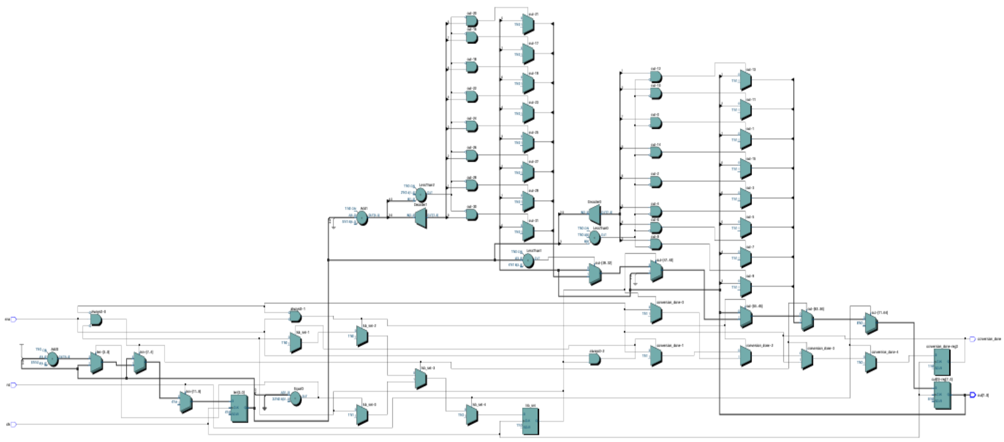

Fig 4

Chapter 6:

Future Trends and Conclusion

6.1 Conclusion

This project successfully demonstrates the FPGA-based implementation of a Successive Approximation Register (SAR) controller using Verilog HDL. Only the SAR block was implemented digitally, while the comparator and DAC were treated as external analog modules. The controller performed the core decision-making in the ADC process by generating trial bits, evaluating comparator feedback, and producing the final digital output through a binary search routine.

The design went through requirement analysis, high-level architecture, Verilog implementation, simulation, synthesis, and hardware testing. Simulation confirmed functional accuracy, while real-time testing validated the controller's ability to work in coordination with actual analog components. The results highlight that FPGAs offer a flexible and powerful platform for implementing custom ADC control logic, providing both educational value and practical applicability in mixed-signal embedded systems.

Overall, the project reinforces key digital design concepts and demonstrates how FPGA-based solutions can enhance the performance and adaptability of analog-to-digital conversion systems.

6.2 Future Trends

Several ongoing developments in ADC technology and FPGA architectures provide opportunities for extending this work:

1. Integration of On-Chip DAC and Comparator

Future FPGA platforms incorporate mixed-signal capabilities. Implementing the entire SAR ADC—including DAC and comparator—on a single chip can significantly reduce noise, latency, and component count.

2. Higher Resolution SAR Controllers

The current design can be scaled to higher resolutions such as 10-bit or 12-bit by modifying the controller logic. This enables applications requiring higher precision like bio-signal monitoring and instrumentation.

3. Low-Power SAR ADC Designs

Emerging trends focus on optimizing power consumption. Techniques such as asynchronous SAR logic and dynamic clocking can be integrated into the controller for battery-operated systems.

4. Machine Learning-Based Calibration

AI-based calibration systems can be integrated with the SAR controller to compensate for comparator offset, DAC nonlinearity, and temperature drift—

enhancing overall ADC performance.

5. Full Mixed-Signal FPGA Implementations

Modern FPGAs offer analog front-end (AFE) blocks. Future work may involve implementing complete ADC architectures like Sigma-Delta or Flash ADCs using a combination of analog and digital FPGA resources.

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[10] Kanphade et al. (2011) – Spartan-II FPGA Implementation of 14-Bit, 20 MS/s, 85 mW SAR ADC suitable for RF Applications. They generated Verilog code via MATLAB/Simulink and synthesized it on Xilinx Spartan II, demonstrating an ADC design with moderate resolution and low power consumption implemented entirely within an FPGA environment.

[11] Ma et al. (2021) – Design and Evaluation of a FPGA-ADC Prototype for PET Detectors based on LYSO Crystals and SiPM arrays. They used a carry-chain Time-to-Digital Converter (TDC) on a Kintex-7 FPGA with just one external resistor, proposing a cost effective integrated solution for PET scanning hardware.

[12] Dhanabalan & Murugan (2020) – FPGA Design of SAR-Type ADC Based Analog Input Module for Industrial Applications. This work developed a SAR-type ADC module for industrial environments, implemented on FPGA and described in a publication in the Advances in VLSI and Embedded Systems

volume.

[13] MDPI (2022) – Design and Implementation of Scalable and Parametrizable Analog-to Digital Converter on FPGA. This paper explores a novel PWM-based ADC on a Xilinx Zynq-7000 FPGA, using DDR-PWM, DDC-PWM, and paralleled outputs to enhance sampling frequency without extra analog components.

[14] Waters (2015) – Automated Verilog-to-Layout Synthesis of ADCs Using Custom Analog Cells (PhD dissertation, Oregon State University). Demonstrates an automated mixed-signal flow where Verilog, combined with custom analog cells, is used to synthesize two ADC types (MASH and pipelined) in CMOS, showcasing rapid and portable mixed-signal design.

[15] MathWorks Technical Article – Designing a Sigma-Delta ADC from Behavioral Model to Verilog and VHDL. This guide explains using Simulink HDL Coder to generate Verilog/VHDL implementations of sigma-delta ADCs from behavioral models, bridging system-level modeling with hardware synthesis.

APPENDIX A:

Plagiarism Report

PLAGIARISM SCAN REPORT

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APPENDIX B:

This appendix contains important segments of Verilog HDL code used in the implementation of the SAR controller. Only essential and representative code blocks are included for clarity.

```
module sar_counter(
   input ena, clk, rst,
   output reg [7:0] out,
   output reg conversion_done
);

   reg [3:0] len;      // Bit index (7 down to 0)
   reg trial_mode;     // 1: Trial, 0: Decision
   reg last_decision_cycle; // Done flag delay

   always @(posedge clk) begin
      if (rst) begin
         out <= 8'b0;
         len <= 7;             // Start MSB
         conversion_done <= 0;
         trial_mode <= 1'b1;    // Start in trial
         last_decision_cycle <= 0;
      end
      else begin
         if (last_decision_cycle) begin
            conversion_done <= 1'b1; // Assert Done
            last_decision_cycle <= 0;
         end
         else if (!conversion_done) begin

            if (trial_mode) begin
               out[len] <= 1'b1; // Set trial bit
               trial_mode <= 1'b0; // Next is decision
            end
            else begin
               if (!ena) begin
                  out[len] <= 1'b0; // Clear bit (Trial too high)
               end

               if (len == 0) begin
                  last_decision_cycle <= 1'b1; // LSB decision made
               end
               else begin
                  len <= len - 1; // Move to next bit
                  trial_mode <= 1'b1; // Next is trial
               end
            end
         end
      end
   end
endmodule
```

APPENDIX C:

Research Publications and Related Work

This appendix summarizes key research papers, technical articles, and reference works that helped shape the development of the SAR controller implemented in this project. These publications provide insight into different ADC architectures, FPGA-based implementations, and digital design techniques that support the theoretical and practical choices made during this work.

C.1 Research Background

Analog-to-Digital Converters (ADCs) are essential components in modern electronic systems, converting real-world analog signals into digital representations for processing, storage, and communication. SAR ADCs in particular have gained widespread use because they offer a good balance between speed, power consumption, and design simplicity. Their digital control logic makes them ideal for implementation on programmable logic devices such as FPGAs.

Several studies have explored both analog and digital aspects of SAR ADCs, as well as the integration of ADCs with programmable hardware. These research works collectively demonstrate the evolution of mixed-signal system design and the increasing role of digital platforms in ADC development.

C.2 Key Publications Reviewed

Homulle et al. (2015)

The authors demonstrated that certain high-speed ADC functions can be implemented directly inside an FPGA using Time-to-Digital Converters (TDCs). Their research highlights the flexibility of FPGAs for mixed-signal experimentation and inspired the digital-only approach used for the SAR controller in this project.

Mindykowski et al. (2012)

This work describes an ADC acquisition setup where a SAR ADC is interfaced with an FPGA before passing data to a DSP processor. Their method illustrates practical challenges in timing, interfacing, and data transfer—issues that are directly relevant to the FPGA–DAC–comparator interfacing done in this project.

Girija et al. (2025)

This study presents the modeling of a SAR ADC using Verilog-A, demonstrating how behavioral descriptions can be used to validate ADC designs. Although this project implements only the digital SAR controller, the paper provided useful insight into the overall structure and operation of SAR ADCs.

Other related works

Additional publications reviewed include studies on low-power SAR ADCs, high-resolution designs, switching methods, and mixed-signal FPGA architectures. These references helped guide decisions related to algorithm flow, bit decision logic, and system-level architecture.

C.3 Relevance to This Project

The research collectively shows that:

- FPGA-based implementation of ADC components is both feasible and educational
- SAR ADCs are among the simplest and most effective architectures for such projects
- A modular design—where the digital SAR controller runs on FPGA and analog components remain external—is widely accepted in academic and industrial prototypes
- Interfacing the digital logic with analog components (DAC and comparator) is a key skill in mixed-signal design

This project follows these trends by implementing only the SAR controller digitally, while using ready-made analog blocks. The design approach aligns with the methods seen in the literature, reinforcing the correctness and relevance of this work.

C.4 Summary

The reviewed publications provided valuable understanding of SAR ADC behavior, FPGA interfacing techniques, and modern mixed-signal design practices. They helped guide the development of the SAR controller logic, the simulation methodology, and the hardware integration strategy used in this mini-project.

Overall, the literature confirms that the chosen approach—FPGA-based SAR controller with external analog components—is both technically sound and widely used in research and educational environments.

IEEE Paper Format